DRS - Digestion and Absorption of Nutrients I

Carbohydrates and Proteins

Lecturer: Dr Li Yueming
Acknowledgements: Dr Bunmi Ibrahim
Dr Majda Bakali



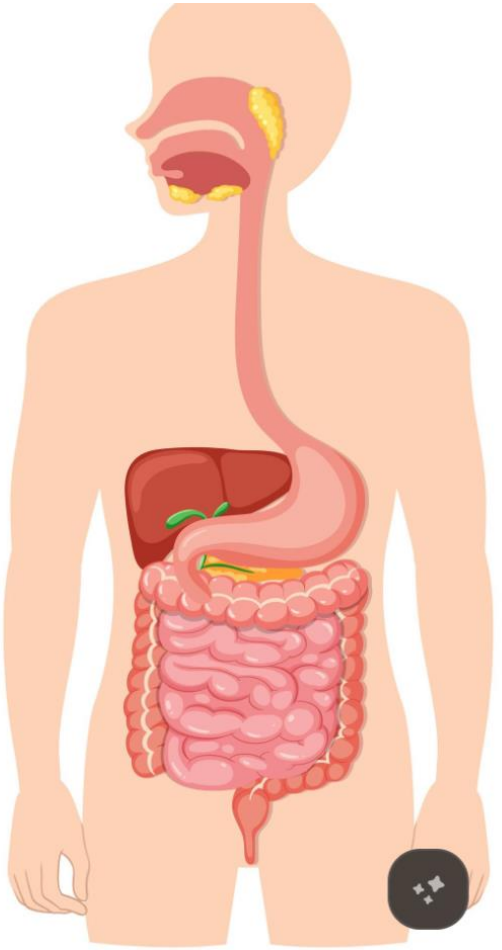
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Learning Outcomes

By the end of this lecture, you should be able to:

- Describe carbohydrate and protein digestion.
- Identify digestive enzymes and their functions.
- Explain carbohydrate and protein absorption.
- Discuss clinically relevant disorders.



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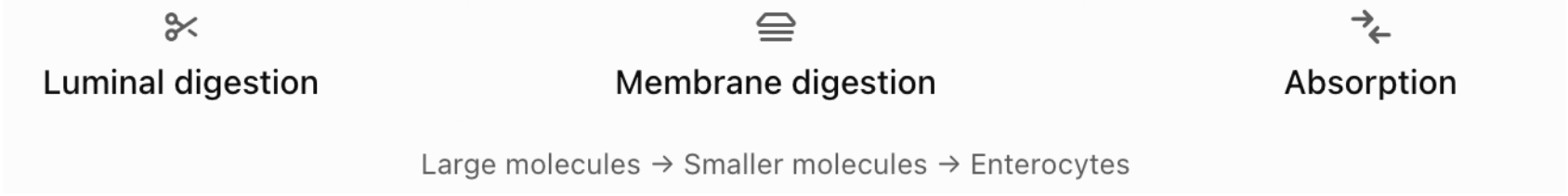
Carbohydrates and proteins: General principle

Carbohydrates and proteins are both polymers

The pathways for the digestion and absorption of carbohydrates and proteins are similar.

Digestion of both polymers is carried out in 2 steps
1. Intraluminal hydrolysis (before the small intestine)
2. Membrane digestion (in the small intestine)

The digested molecules are then absorbed by the enterocytes



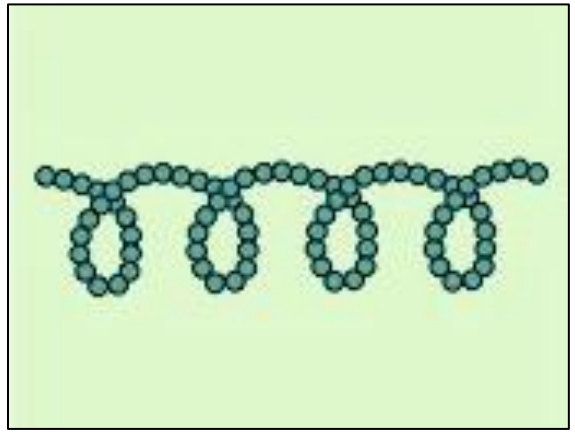
Carbohydrates

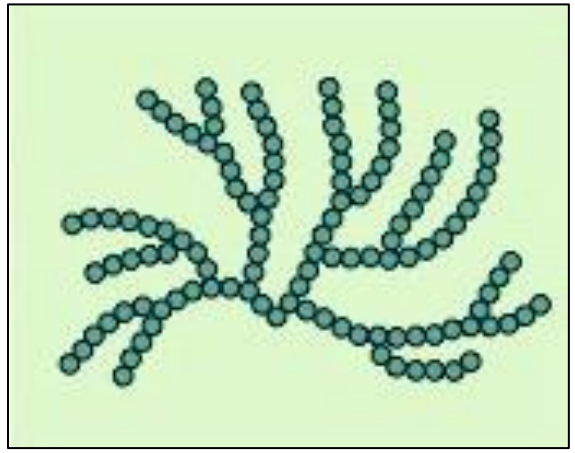


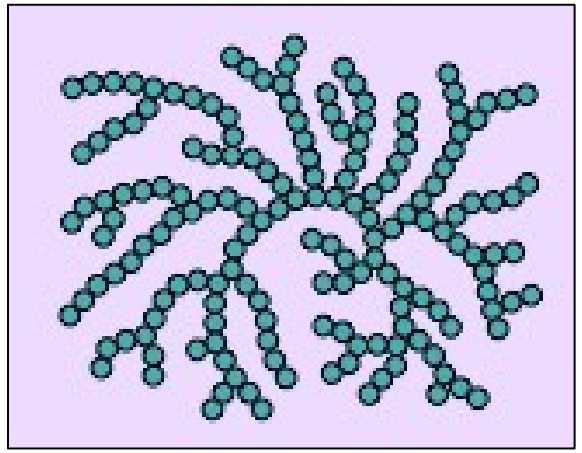
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Carbohydrates

- Starch
  - 45% to 60% of dietary carbohydrate is in the form of starch
  - Polysaccharide / Glucose polymer
  - Storage form for carbohydrates
  - Primarily found in plants
  - Consists of both amylose and amylopectin
- Glycogen is a storage form for carbohydrates
  - Animal starch
  - Like amylopectin
  - α-1,4 and α-1,6 linkages but more highly branched



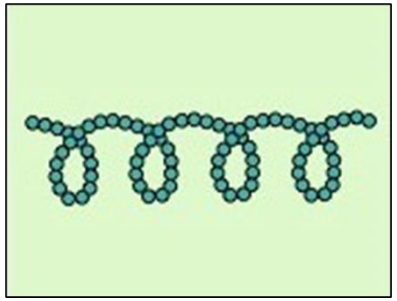




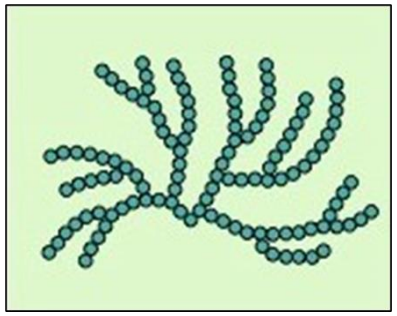
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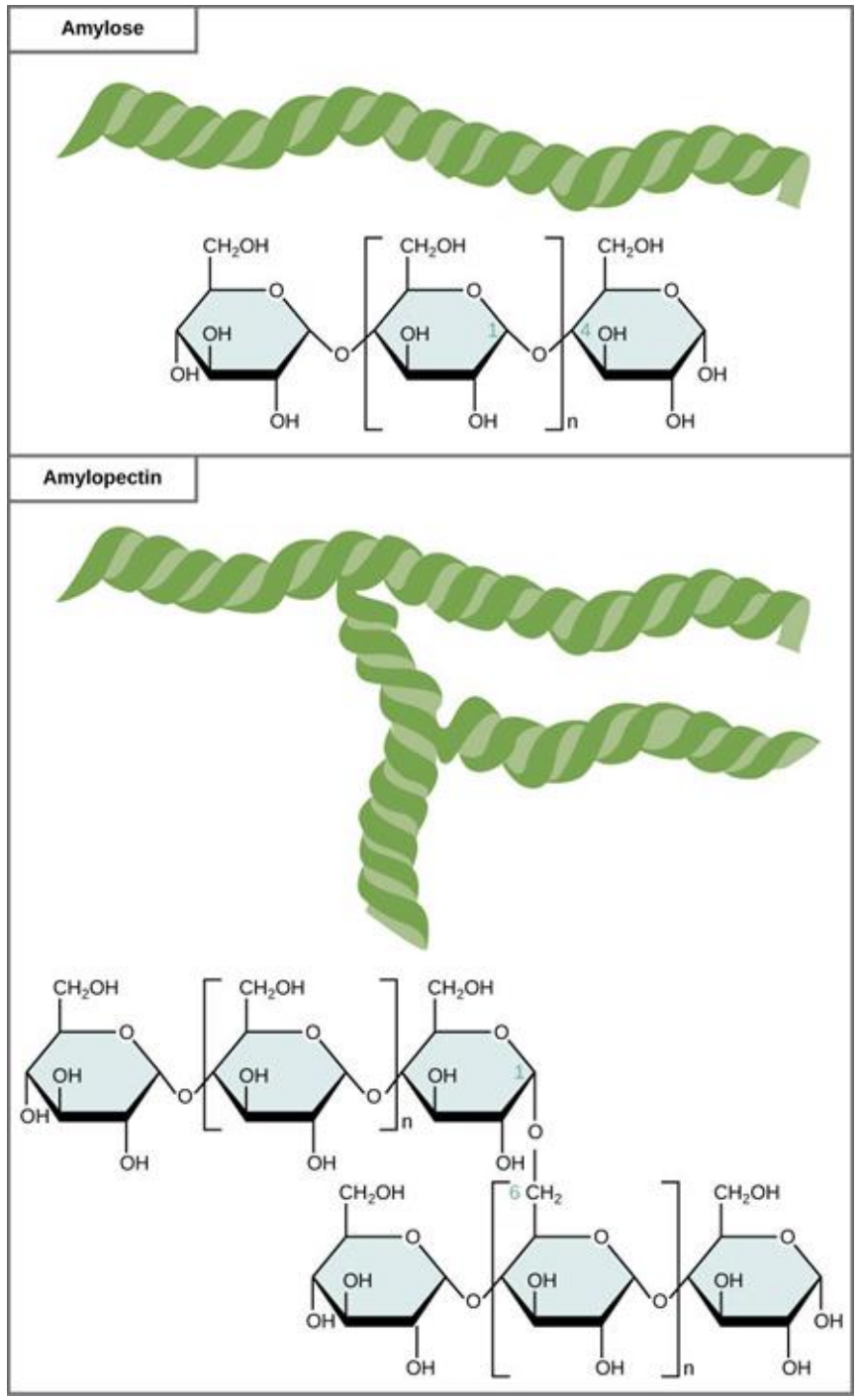
Starch

- Amylose
  - Straight-chain glucose polymer
  - Multiple glucose residues
  - Connected by α-1,4 linkages



- Amylopectin
  - Branched glucose polymer
  - Contains both α-1,4 and α-1,6 linkages





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Carbohydrates



- 30% to 40% of dietary carbohydrates are disaccharides
  - **Sucrose** derived from sugar cane and sugar beets
  - **Lactose** is the sugar found in milk
- The remaining 5% to 10% are monosaccharides
  - Fructose
  - Glucose





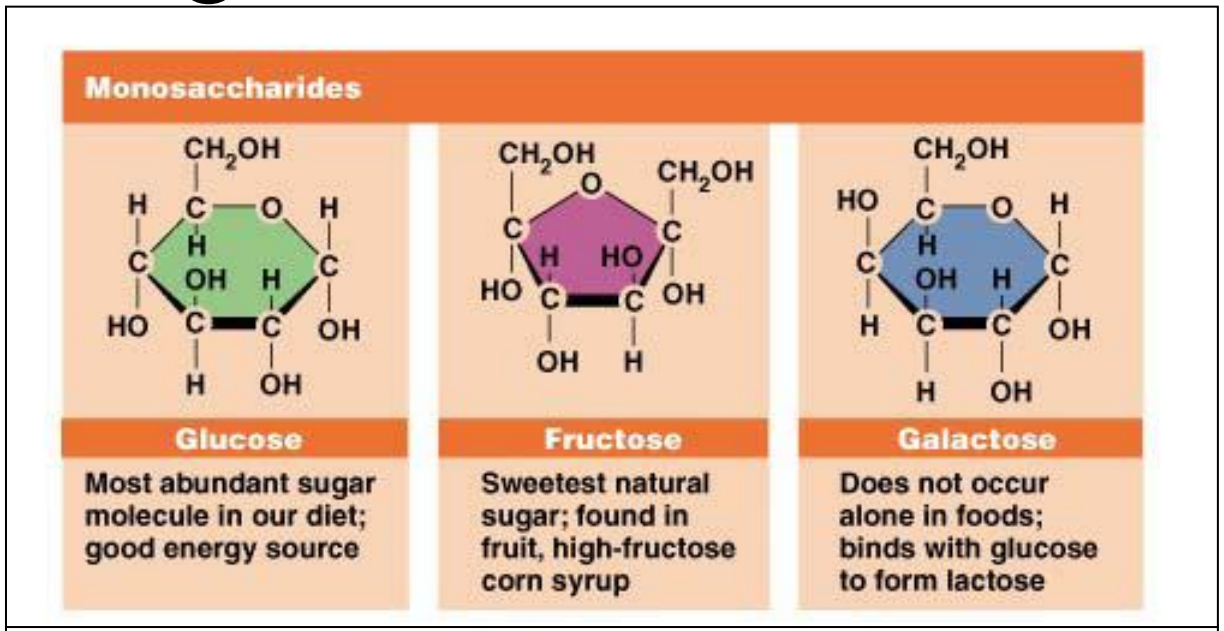
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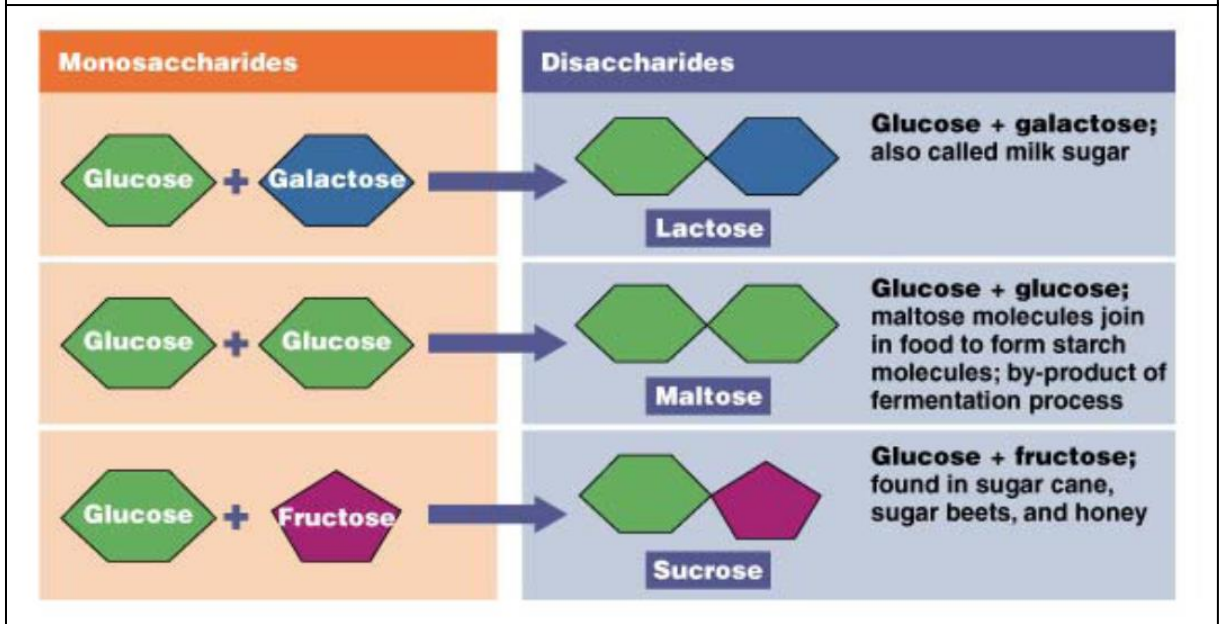
Why Must Carbohydrates Be Digested?

Only monosaccharides can be absorbed by the small intestine.

- There is no evidence of any intestinal absorption of either starches or disaccharides
- Digestion is required before absorption.

Monosaccharides: Glucose, fructose, galactose.





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Two Stages of Carbohydrate Digestion

1. Intraluminal digestion

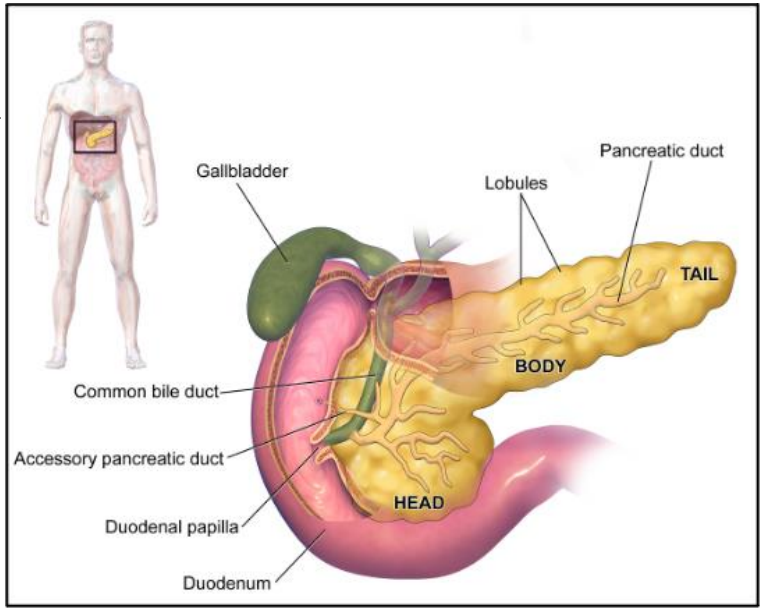
Starch → Oligosaccharides

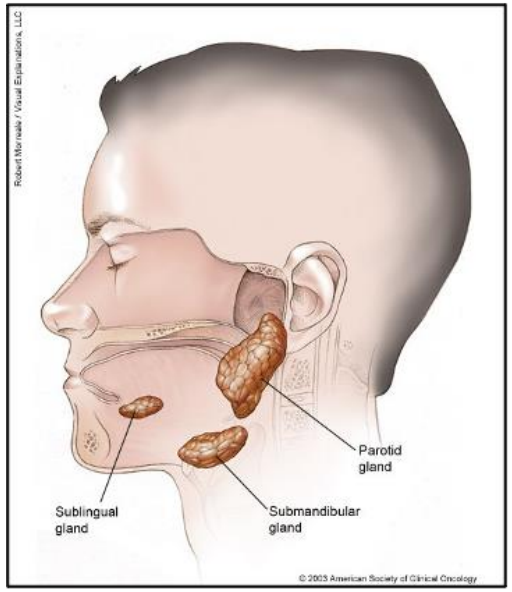
- Salivary α-amylase
- Pancreatic α-amylase

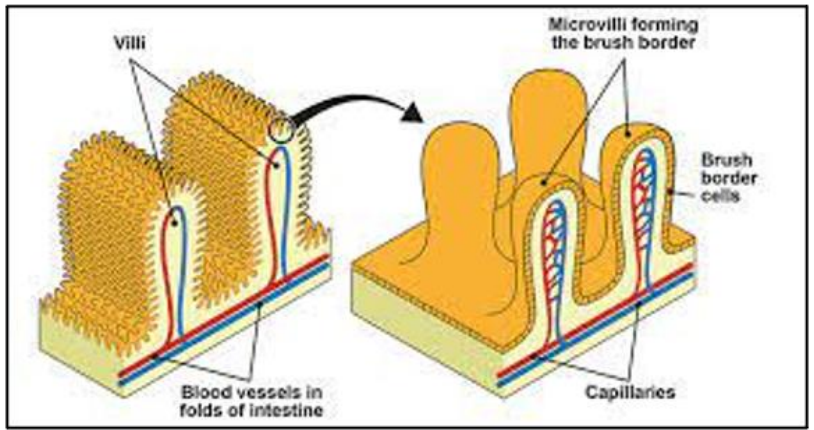
2. Membrane digestion

Oligosaccharides → Monosaccharides

- Brush-border disaccharidases







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Enzymes Involved in Carbohydrate Digestion

Enzyme	Location	Substrate	Type of digestion
α-amylase	Saliva Pancreas	Polysaccharides	Luminal
Disaccharidases	Intestine	Disaccharides	Membrane
α-1,6-glucosidase Isomaltase	Intestine	Branch points of oligo- and disaccharides	Membrane

- Salivary and pancreatic α-amylase are present in serum.
- Salivary amylase is inactivated by stomach acid.
- Pancreatic amylase plays the major role.
- Increased serum pancreatic amylase is associated with acute pancreatitis.

1 - Luminal Digestion

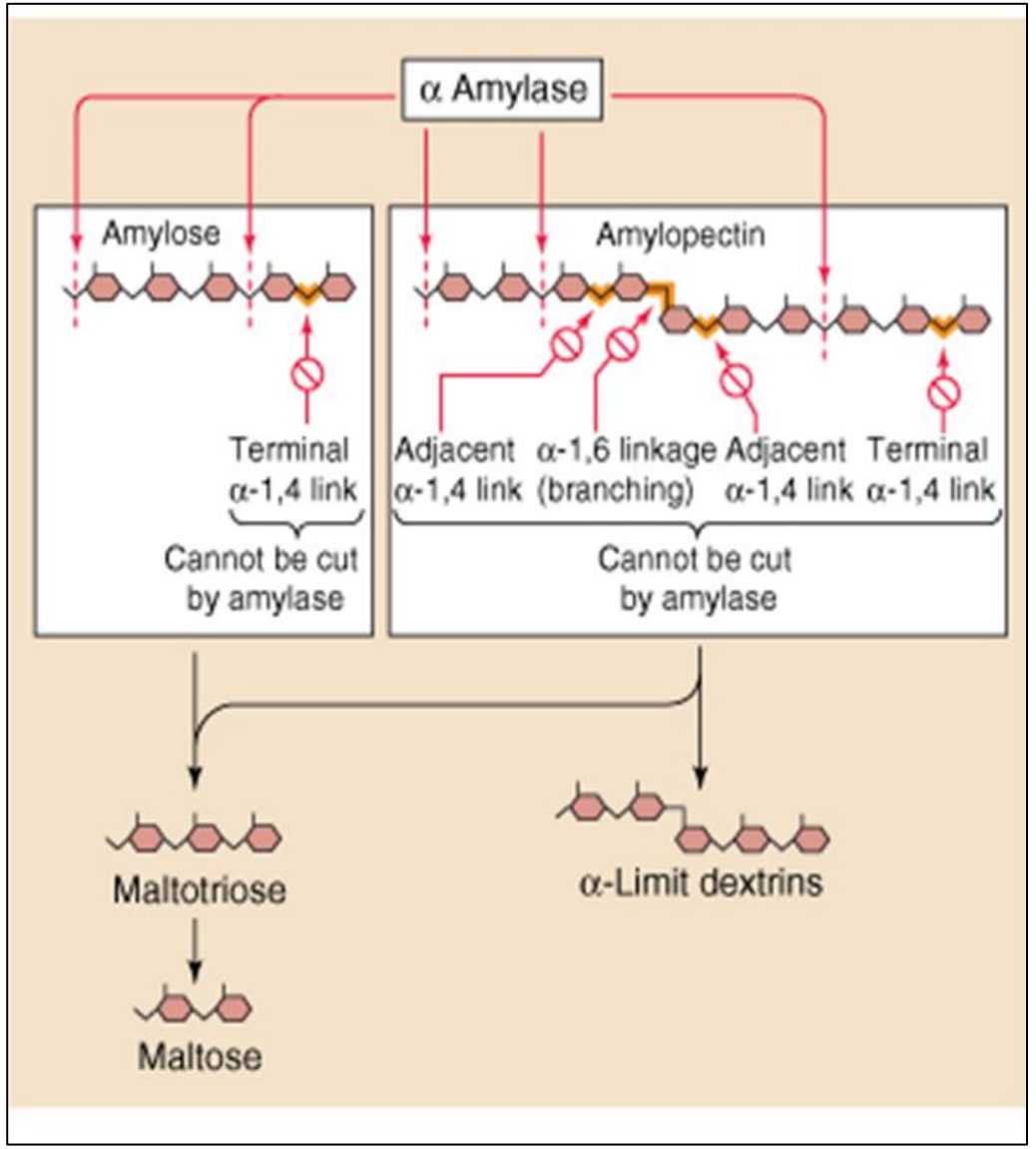
- Amylase is an endoenzyme that cleaves **α-1,4 bonds**

1. Salivary Amylase
   - Starts the process > Inactivated by acid
2. Pancreatic Amylase
   - Induced by CCK
   - Completes starch digestion

- Amylase cannot act on the following
  - Terminal α-1,4 links or α-1,6 links
  - α-1,4 links adjacent to α-1,6 links

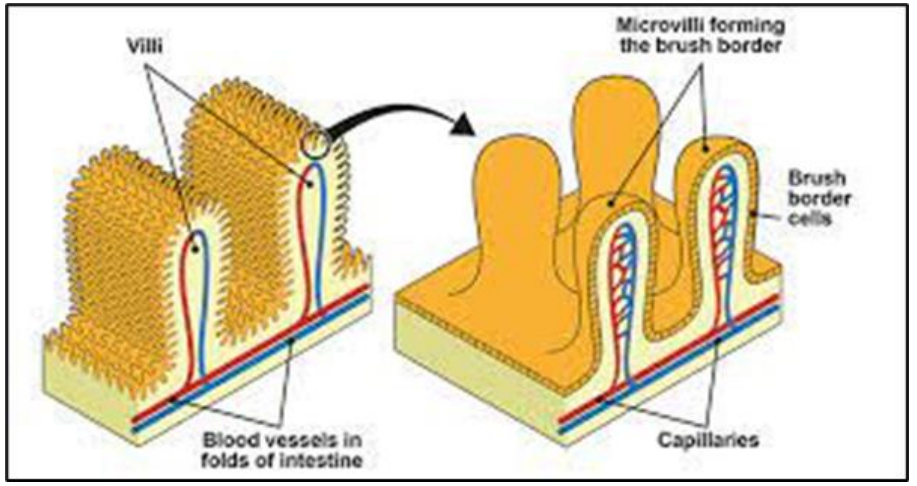
Products: Maltose, maltotriose, α-limit dextrins.

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- Three products: Further breakdown required !!!

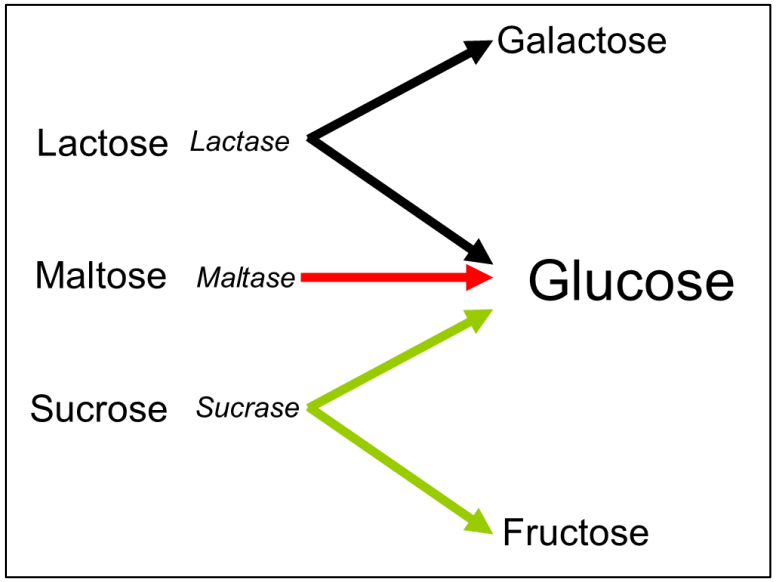
Membrane Digestion: Brush-Border Enzymes



Brush-border disaccharidases

Enzyme	Function
Maltase	Maltose → Glucose
Sucrase	Sucrose → Glucose + Fructose
Lactase	Lactose → Glucose + Galactose
Isomaltase	Cleaves α-1,6 bonds

- Maltase, sucrase and isomaltase: Cleavage of α-1,4 linkages.
- Membrane digestion produces absorbable monosaccharides.



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Carbohydrate Absorption: SGLT1, GLUT5 and GLUT2

Monosaccharide Transport

SGLT1

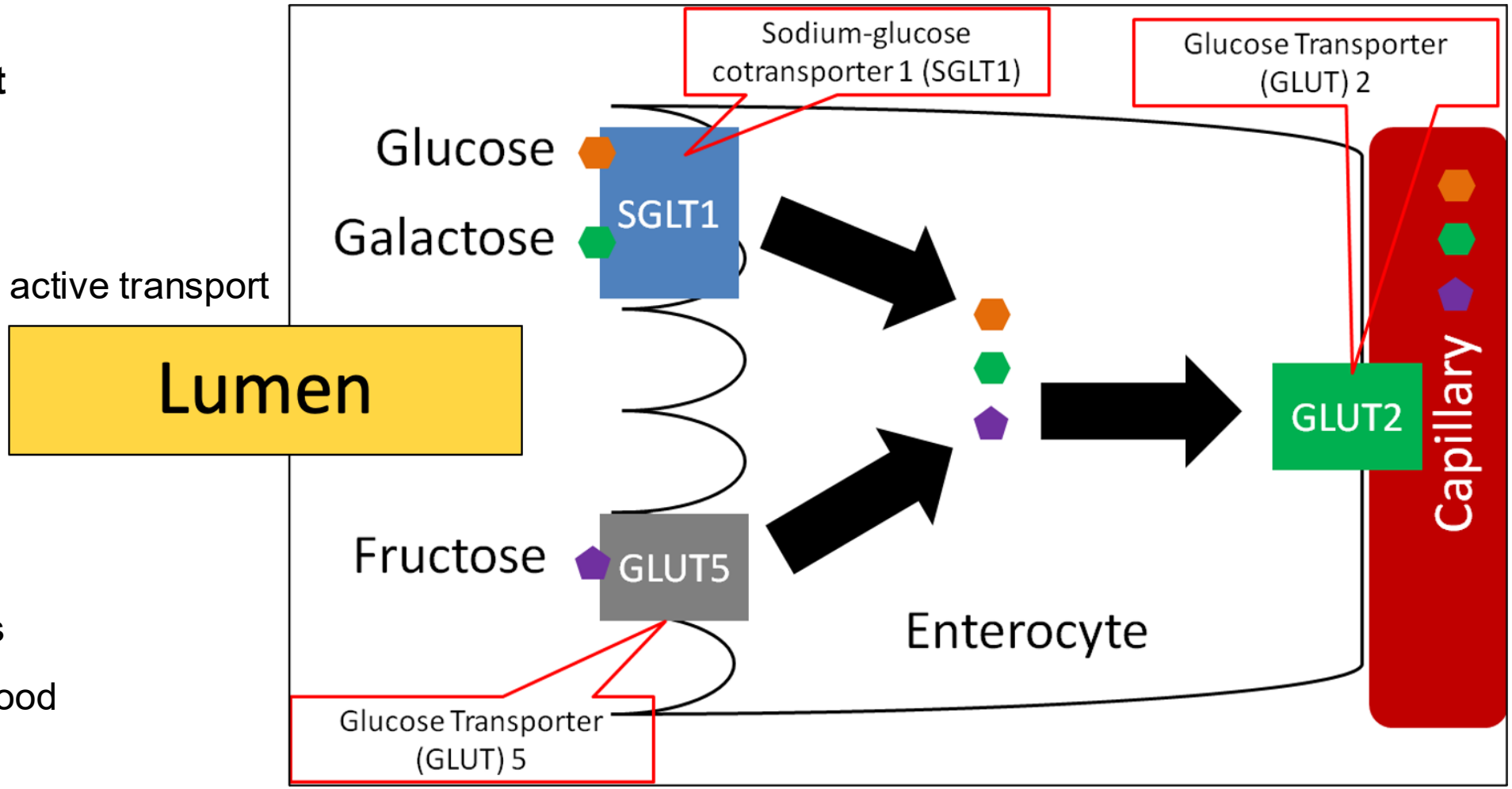
- Glucose and galactose
- Na^+-dependent secondary active transport

GLUT5

- Fructose
- Facilitated diffusion

GLUT2

- All three monosaccharides
- Facilitated diffusion into blood

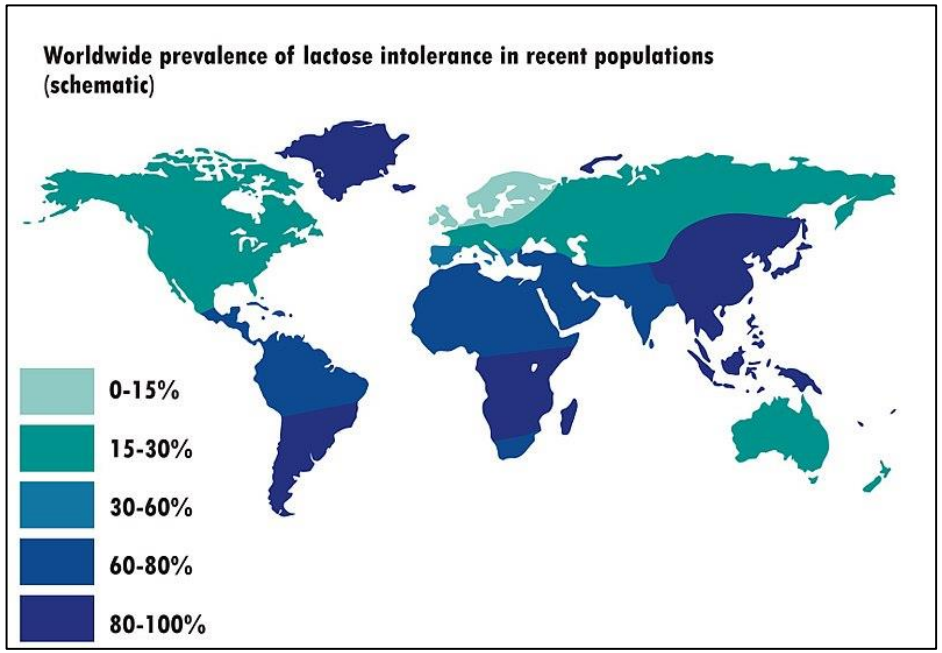


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Clinical relevance of carbohydrate digestion

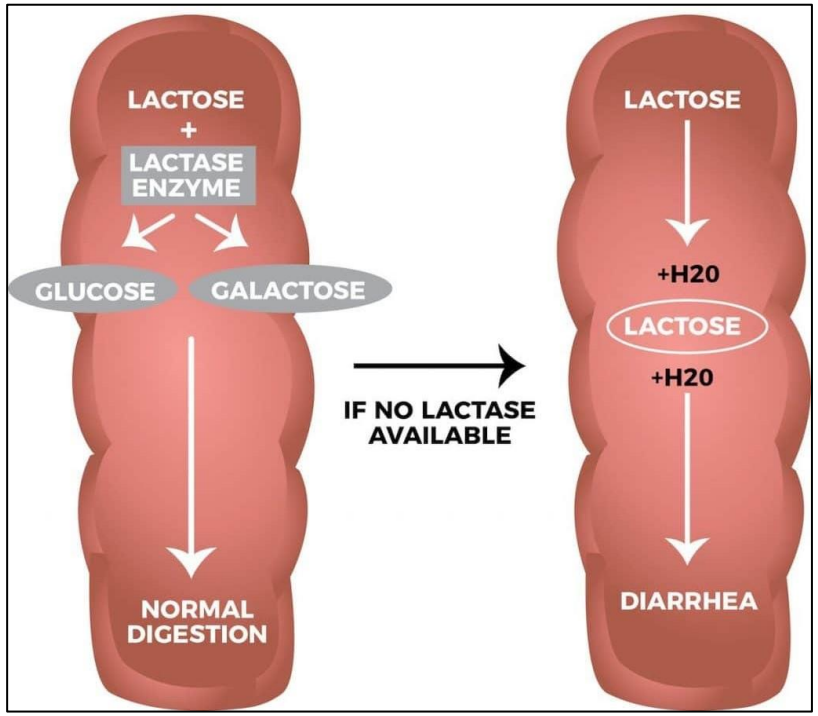
Lactase deficiency

- Lactase activity decreases after weaning.
- Lactose digestion is reduced.
- Prevalence varies between populations.



Symptoms

- Abdominal cramps
- Bloating
- Osmotic diarrhoea



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Silanikove *et al.*, 2015. doi: 10.3390/nu7095340
Stock & Wells, 2023. doi: 10.1093/af/vfad022

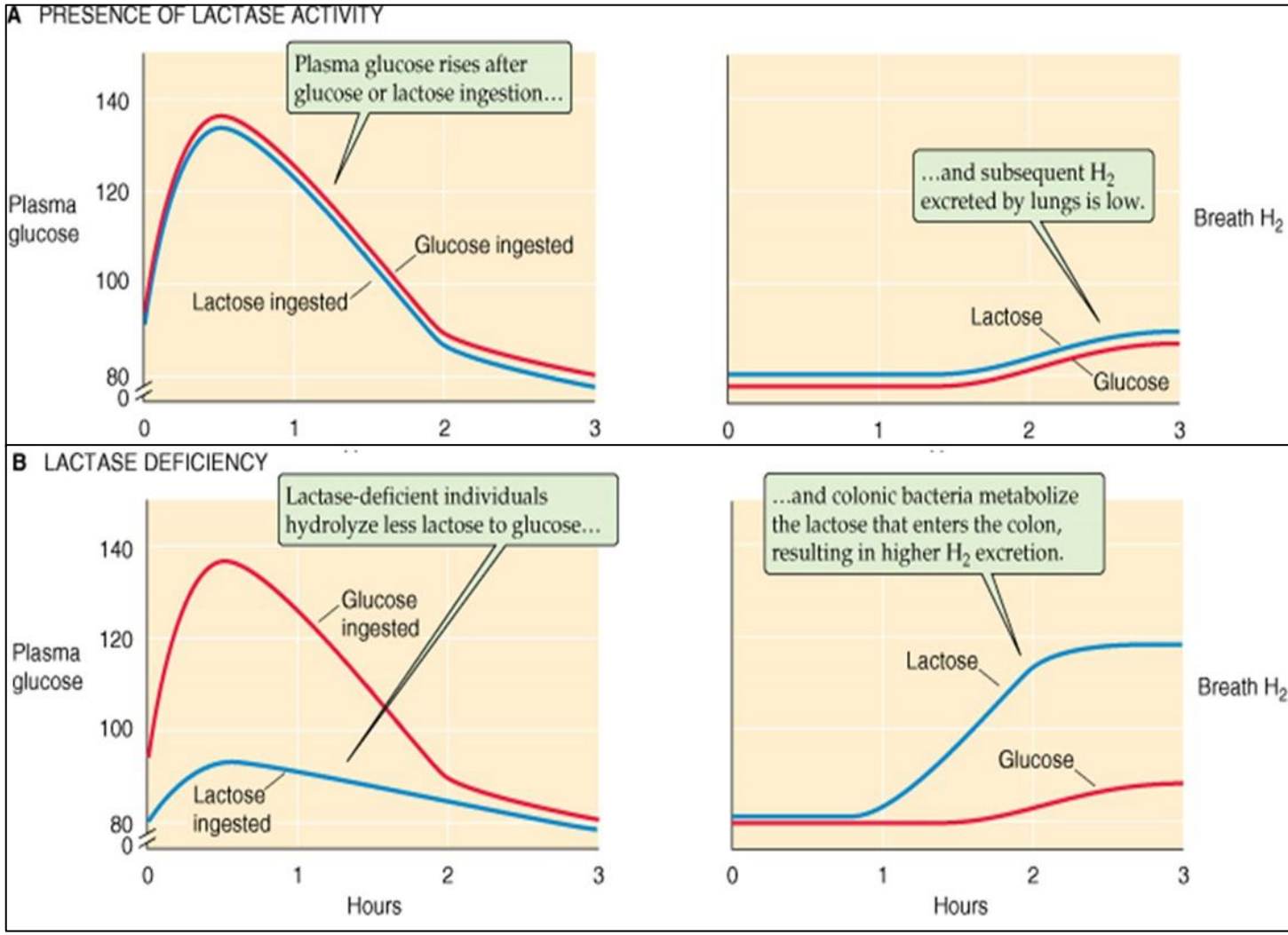
Clinical relevance of carbohydrate digestion

Lactase Deficiency: Clinical Measurements

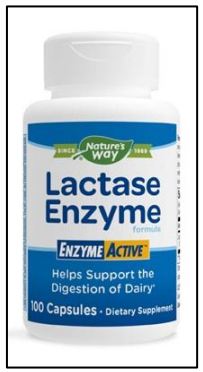
Normal lactase activity
- Plasma glucose increases after lactose ingestion.
- Breath hydrogen remains low.

Lactase deficiency
- Smaller increase in plasma glucose.
- Increased breath hydrogen after lactose ingestion.

Examples : Lactase supplements and lactose-free milk.



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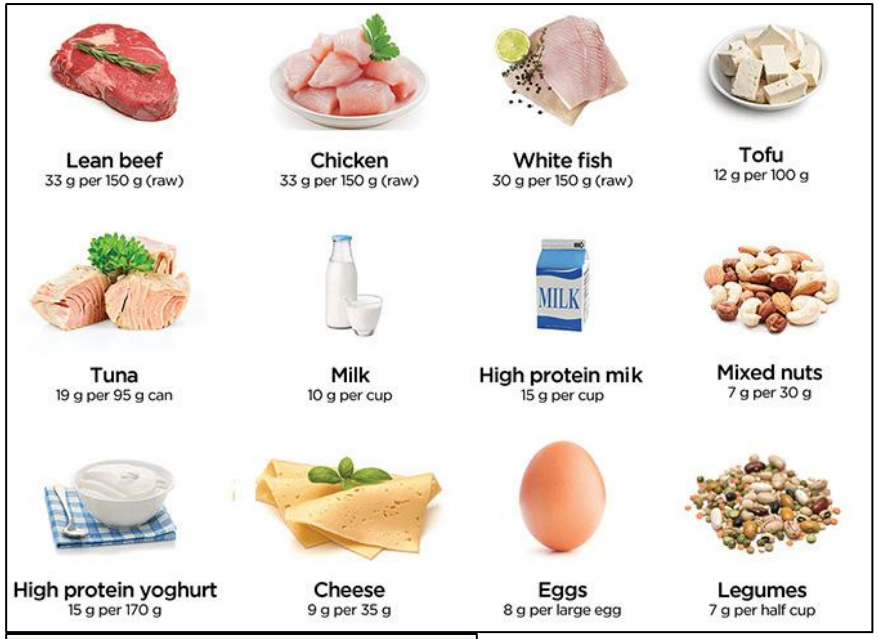


Proteins

Protein Sources and Digestion

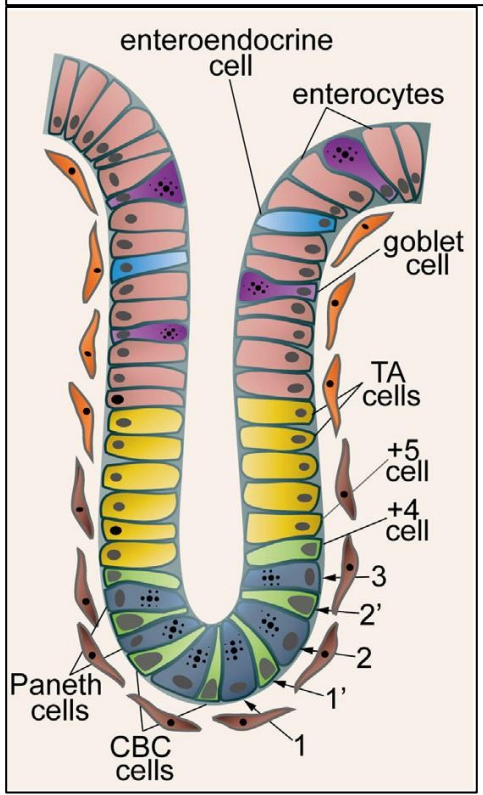
Sources of proteins in the digestive tract

50%	50%
Dietary proteins	Endogenous proteins
	Enzymes, hormones, desquamated cells



Digestion and absorption
Proteins → Oligopeptides + Amino acids → Enterocytes

Exception in neonates: Intact immunoglobulins from breast milk may be taken up by endocytosis.



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Protein Digestion in the Stomach

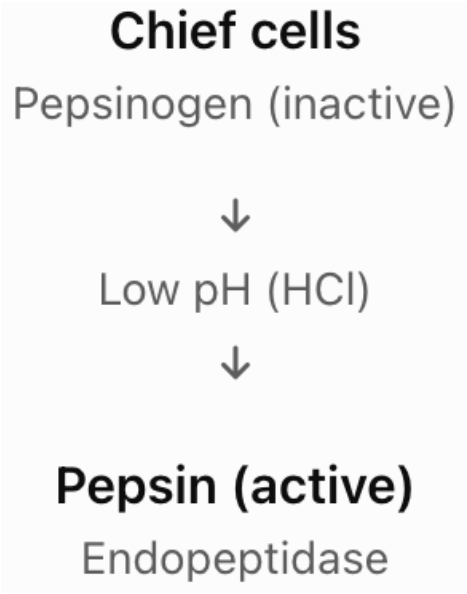
- Process for luminal digestion involves gastric and pancreatic **proteases**
  - Secreted as pro-enzymes

Chief cells

- Pepsinogen
- Low pH (HCl) > Pepsin

Pepsin

- Endopeptidase (aromatic AA's – TYR & PHE)
- Partially digests 10-15% of protein



Pepsinogen $\xrightarrow{HCl}$ Pepsin

Dietary proteins $\xrightarrow{Pepsin}$ Large polypeptides

Pepsin is an endopeptidase with primary specificity for peptide linkages of aromatic and larger neutral amino acids.

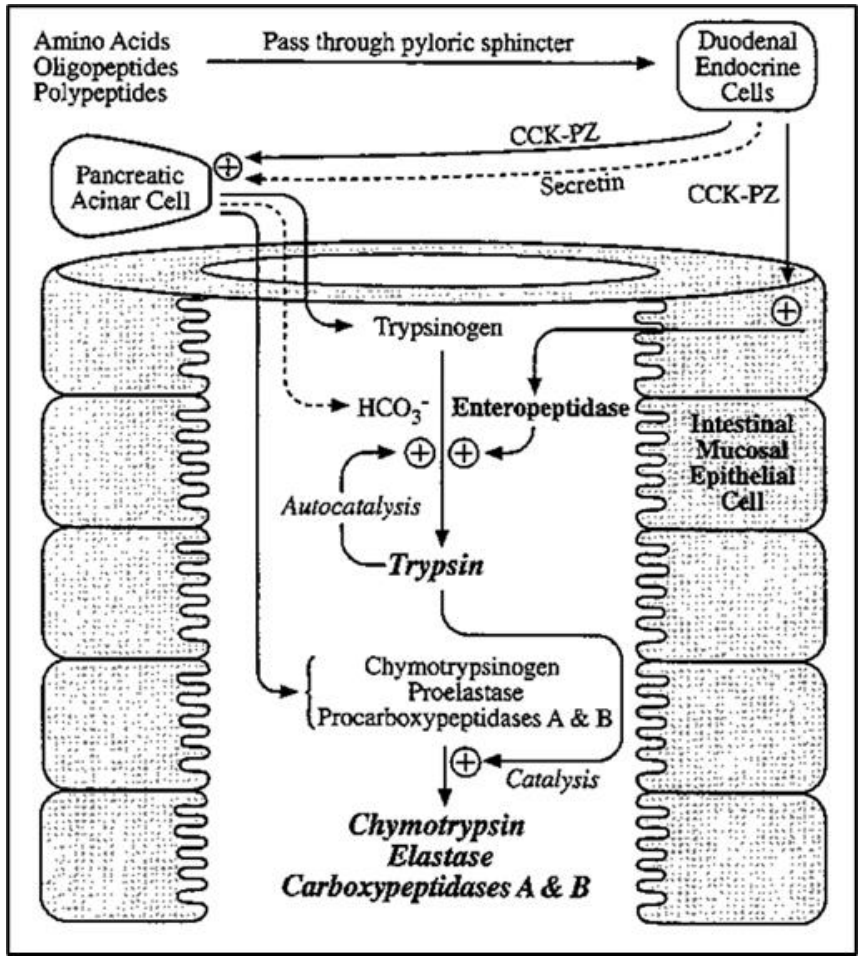
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Activation of Pancreatic Proteases

Pancreatic Proteases

- Produced by pancreatic acinar cells
- Secreted as inactive proenzymes
- Duodenum: main site of protein digestion

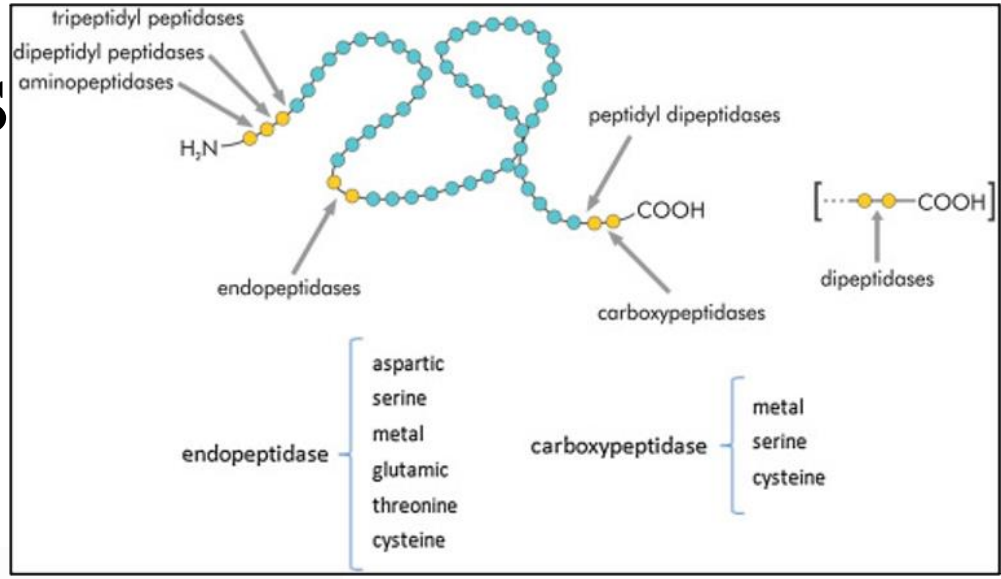
Proenzyme	Activating agent	Active form
Trypsinogen	Enterokinase	Trypsin
Chymotrypsinogen	Trypsin	Chymotrpysin
Proelastase	Trypsin	Elastase
Procarboxypeptidase A	Trypsin	Carboxypeptidase A
Procarboxypeptidase B	Trypsin	Carboxypeptidase B



CCK-PZ is CCK and Enteropeptidase is enterokinase

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Two types of pancreatic proteases



Endopeptidases vs Exopeptidases

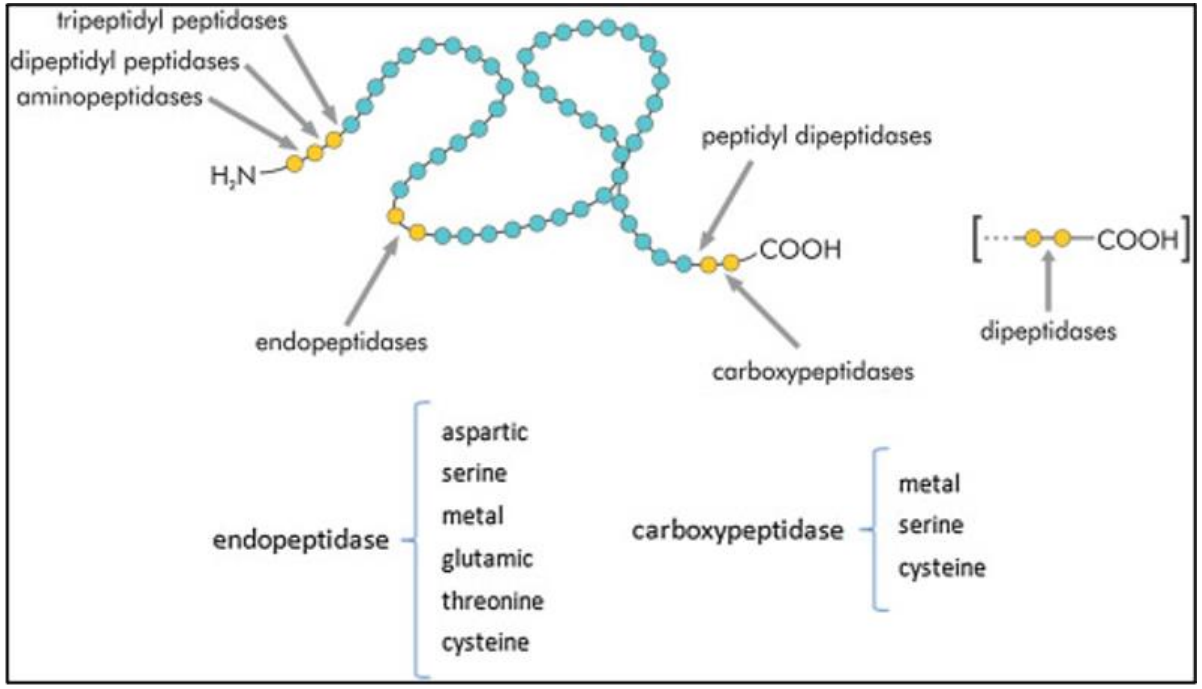
	Endopeptidases	Exopeptidases
Enzymes	Trypsin, chymotrypsin, elastase	Carboxypeptidases A & B
Action	Cleave internal peptide bonds	Cleave bonds near C-terminus
Polypeptides	~70%	~30%
Products	Oligopeptides (2–6 amino acids)	Individual amino acids

Membrane Digestion of Proteins

Brush-Border and Intracellular Peptidases

- Dipeptidase
- Aminopeptidase
- Tripeptidyl peptidase
- Dipeptidyl peptidase
- Peptidyl dipeptidase

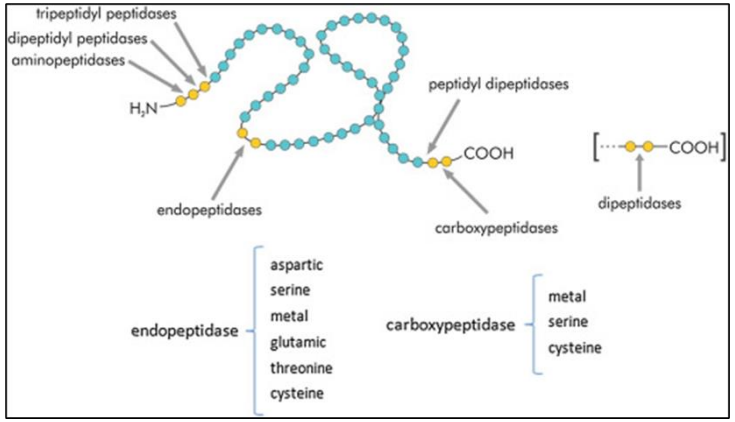
Function: Further hydrolysis of small peptides into amino acids.

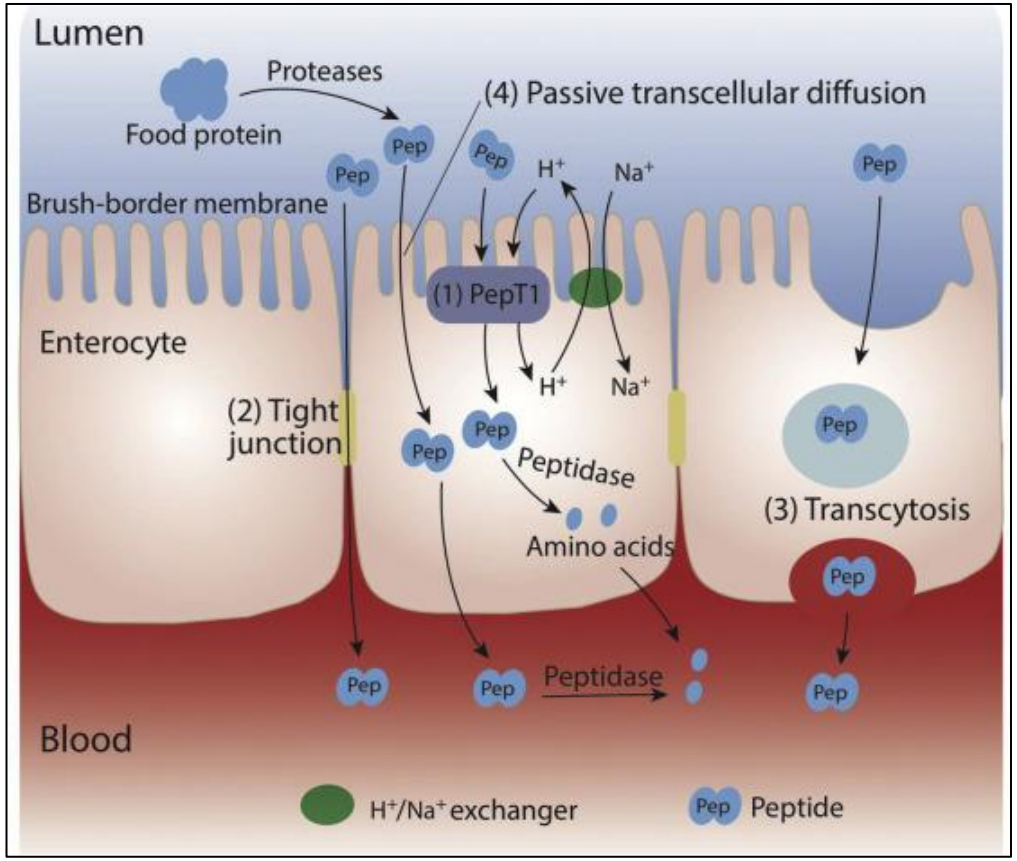


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Absorption

- H^+ dependent co-transporter (PepT1)
  - Tripeptides are absorbed
  - Cellular peptidases further hydrolyse the peptide into amino acids
- H^+-Na^+ co-transporter
  - Dipeptides are absorbed
  - Cellular peptidases further hydrolyse the peptide into amino acids
- Amino acid Na^+
  - Na^+ dependant co-transporter
  - AA's > inside the enterocytes



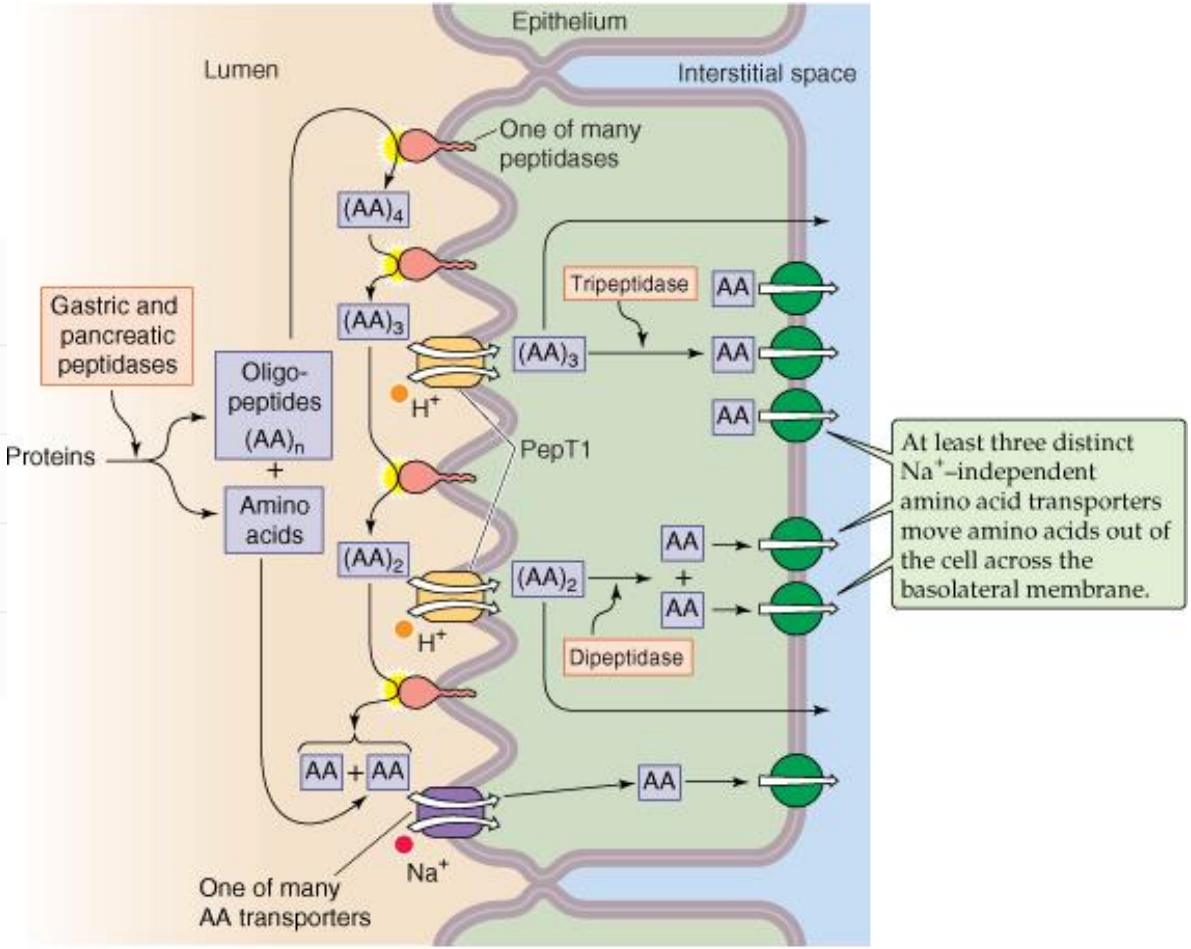


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Amino Acid Transport Systems

Amino Acid Membrane Transporters

Amino acid type	Transport mechanism
Glycine / Proline	IMINO transporter; Na^+ cotransport
Positively charged AA	Antiporter
Neutral AA	Na^+ cotransporter
Negatively charged AA	Na^+ cotransporter (EAAT3)



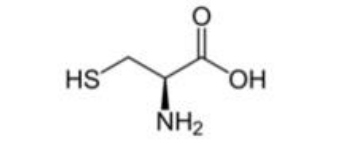
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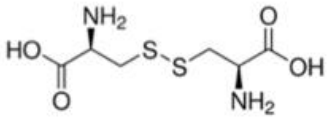
Clinical Relevance: Cystinuria

Cystinuria

- Autosomal recessive hereditary disorder
- Mutations affecting renal amino acid transporters
- Impaired cystine reabsorption in the proximal tubule

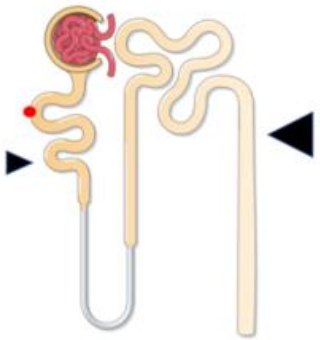


Cysteine



Cystine

Cystinuria pathophysiology





Recurrent nephrolithiasis
Renal colic
Acute kidney injury
Chronic kidney disease
Kidney failure

Defective amino acid transport can cause disease.

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Summary: Digestion and Absorption of Nutrients

Carbohydrates

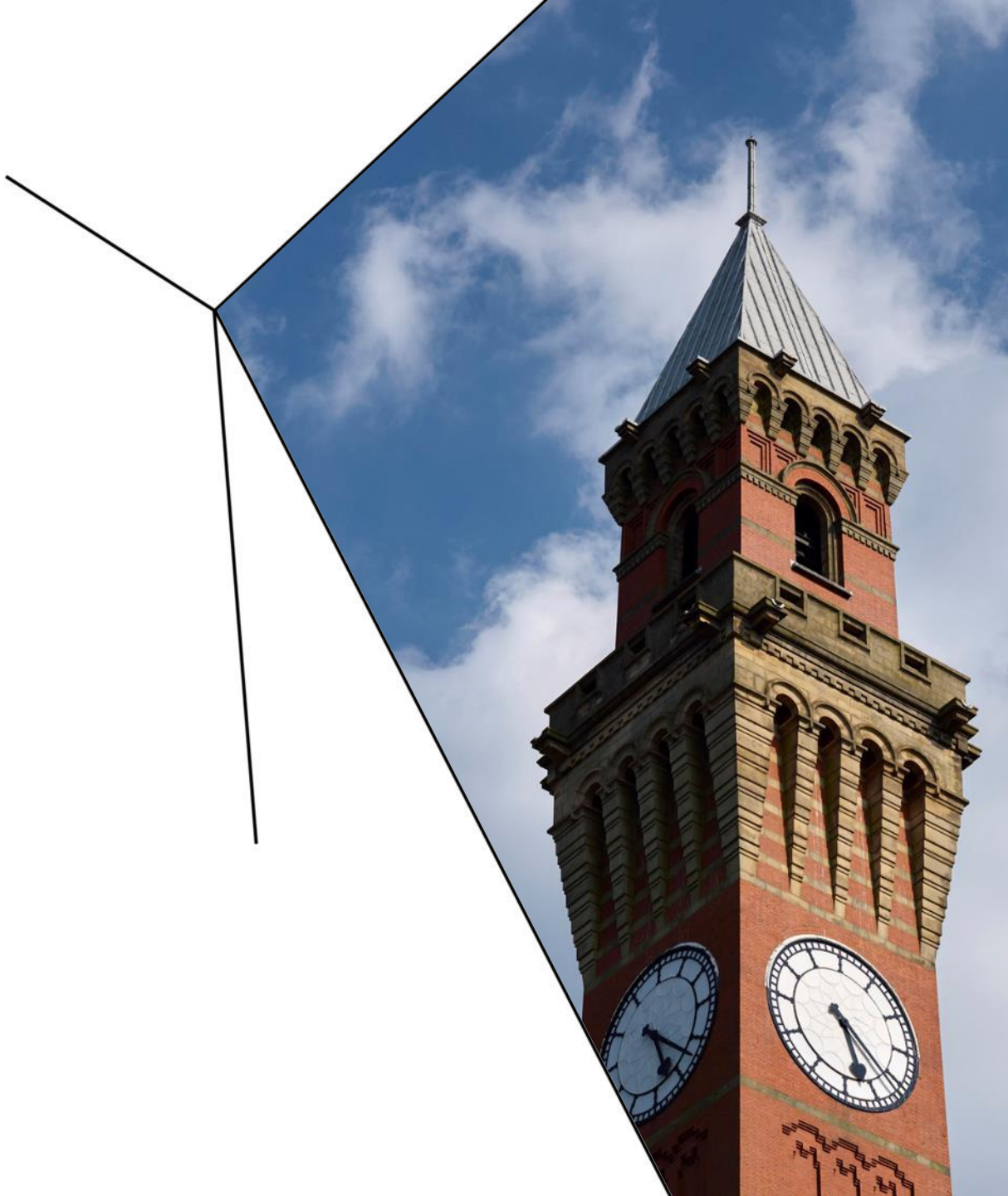
- Luminal digestion: α-amylase
- Membrane digestion: disaccharidases
- Absorption: SGLT1, GLUT5, GLUT2

Clinical relevance: Lactase deficiency

Proteins

- Luminal digestion: pepsin, pancreatic proteases
- Membrane digestion: peptidases
- Absorption: amino acid transporters, PepT1
- **Clinical relevance:** Cystinuria

Questions



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